Logic and Set Theory
Lecture N°3
1 Lic Data Science

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ALGEBRA1

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Set Theory
What is a Set?
subsets
Basic Set Operations
Union
Intersection
COMPLEMENT OF SET
The difference of sets
Generalized union and intersection
Symmetric Difference of Sets
Cartesian Product

Summary

Set Theory

- What is a Set?
- subsets

Basic Set Operations

- Union
- Intersection
- Generalized union and intersection
- Symmetric Difference of Sets
- Cartesian Product

What is a Set?

- A **set** is a well-defined collection of distinct objects.
- The objects in a set are called the **elements** or **members** of the set. The logical statement "a is a member of the set A" is written

$$a \in A.$$

- Lowercase letters $a, b, c, \cdots$ denote the elements of a set.
- Capital letters $A, B, C, \cdots$ usually denote sets.

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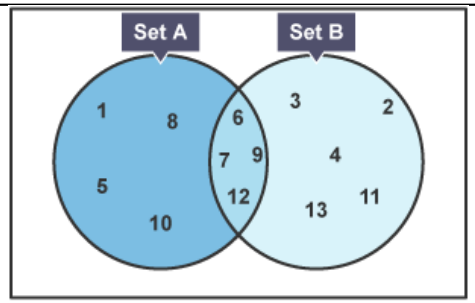
Remark

Likewise, its logical negation "a is not a member of the set A" is written $a \notin A$.
Therefore, exactly one of these two statements is true.

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Representation of sets

There are three main ways to identify a set:

Name	Definition	Example
Roster Method	The set can be defined by listing all its elements, separated by commas and enclosed within braces.	$A = \{0, 2, 4, 6\}$
Set Builder Method	The set can be defined by describing the elements using mathematical statements. $A = \{x \| P(x)\}$	$A = \{x \in \mathbb{N} \| 0 \leq x < 9\}$
Venn Diagram	A Venn Diagram is a pictorial representation of the relationships between sets.	

Example

- $B = \{red, blue, green\}$ is the set of colors.
- $C = \{1, 2, 3, 4, 5\}$ is the set of the first five natural numbers..
- The collection of real numbers that satisfy the equation.
- The collection of two-digit positive integers divisible by 5.
- The collection of great football players in the National Football League.
- The set of students who never forget their homework.

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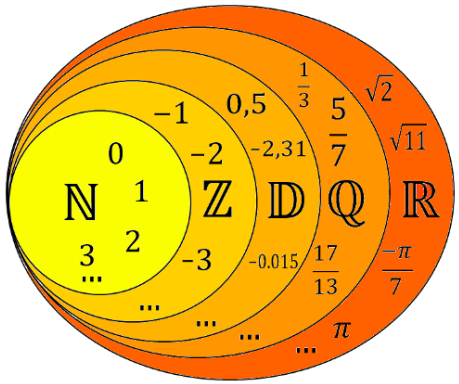
Example

$\mathbb{N}$: *the positive integers* or Natural numbers,
$\mathbb{Z}$: *the integers*,
$\mathbb{Q}$: *the irrational numbers*,
$\mathbb{R}$: *the real numbers*,
$\mathbb{C}$: *the complex numbers*

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Example

$\mathbb{N}$: *the positive integers* or Natural numbers,
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The Empty Set

- The set with no elements.
- Also called the null set.
- Denoted by the symbol $\emptyset$.

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The Empty Set

- The set with no elements.
- Also called the null set.
- Denoted by the symbol $\emptyset$.

Example

The set of real numbers x that satisfy the equation

$$x^2 + 1 = 0.$$

Definition

For a finite set A, the **cardinality** $|A|$ equals the number of elements in A. If there is a bijective mapping between the set A and the natural numbers $\mathbb{N}$, then $|A| = \infty$ and the set is called **countably infinite**. If $|A| = \infty$ and the set is not countably infinite, then A is called **uncountably infinite**.

Example

The set of rational numbers is countably infinite while the set of real numbers is uncountably infinite.

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Remark

i The set with no elements and is called an *empty set*. noted $\emptyset$.

ii A *singleton* set is a set containing exactly one element such as $\{a\}$.

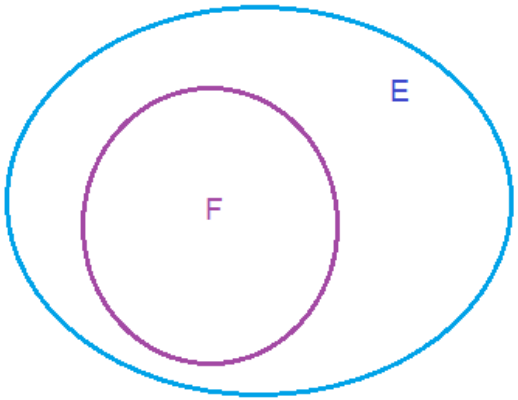
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Definition

Let E and F be sets. We say that F is a **subset** of E iff every element of F is also an element of E. In symbols this is

$$F \subseteq E \Longleftrightarrow (\forall x)((x \in F) \Rightarrow (x \in E)).$$

It is a **proper subset** (denoted $A \subset B$) if $A \subseteq B$ and $A \neq B$



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Theorem

If A is a subset of B and B is a subset of C, then A is a subset of C. That is

$$(A \subseteq B \wedge B \subseteq C) \Rightarrow A \subseteq C.$$

Proof.

To prove that $A \subseteq C$, we must show that
$\forall x(x \in A \Rightarrow x \in C)$.
Since $A \subseteq B$, then
$\forall x(x \in A \Rightarrow x \in B)$,
and since $B \subseteq C$, then
$\forall x(x \in B \Rightarrow x \in C)$.
Hence $\forall x(x \in A \Rightarrow x \in C)$.
Therefore $A \subseteq C$. □

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Definition

For an arbitrary set A, the set of all the subsets of A is called **the power set of A**. We denote the power set of A by $P(A)$. Thus

$$P(A) = \{B : B \subseteq A\}.$$

Example

1. Let $A = \{4, 7\}$. Then

$$P(A) = \{\emptyset, \{4\}, \{7\}, A\}.$$

2. Let $M = \{a, b, c\}$. Then

$$P(M) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, M\}.$$

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Remark

1. The number of elements in the power set of A is 2^n, where n is the number of elements in set A. That is $card(P(A)) = 2^n$, where $n = card(A)$.
2. $B \in P(A)$ means $B \subseteq A$.

The basic Set Operations are:

Symbol	Definition	Name
$A \cup B$	$\{x \mid x \in A \quad or \quad x \in B\}$	union
$A \cap B$	$\{x \mid x \in A \quad and \quad x \in B\}$	intersection
$\overline{A}; A^C$	$\{x \mid x \in E \quad and \quad x \in A\}$	complement
$A \ B$	$\{x \mid x \in A \quad and \quad x \notin B\}$	set difference
$A \Delta B$	$A \ B \cup B \ A$	set difference

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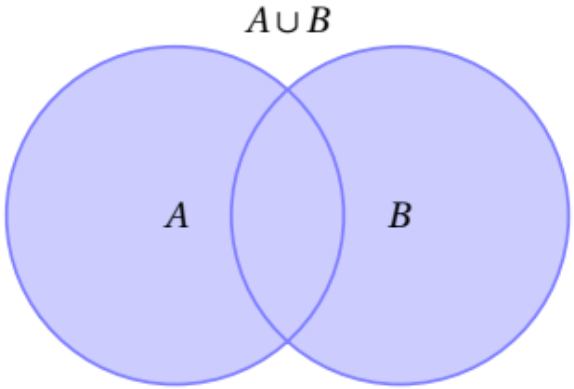
Union

Definition

If A and B are sets. The union of A and B, denoted by $A \cup B$, is the set defined by

$$A \cup B = \{x \in E \mid (x \in A) \vee (x \in B)\}$$

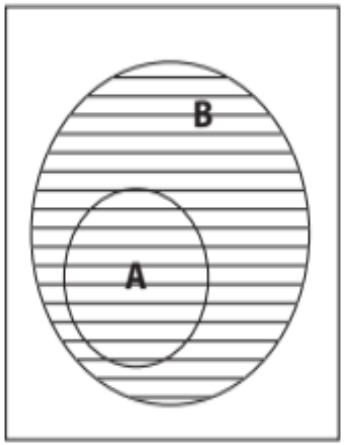
That is, $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$.



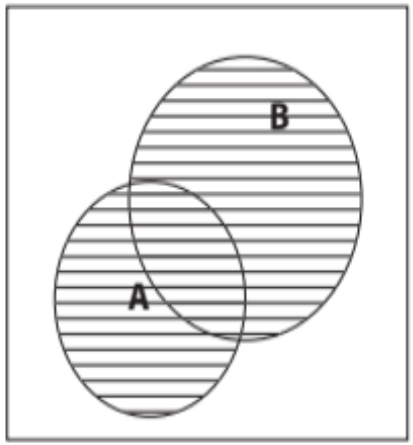
Example

lets $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 9\}$.Then

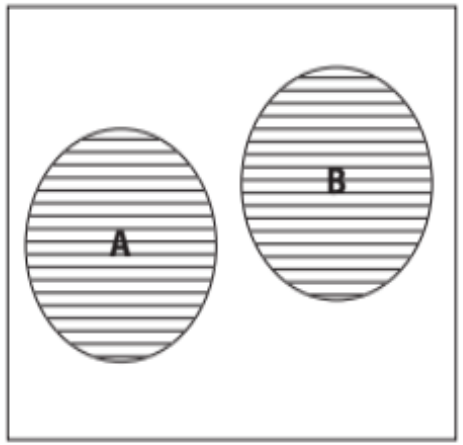
$$A \cup B = \{1, 2, 3, 4, 6, 9\}.$$



$A \cup B$ is shaded



$A \cup B$ is shaded



$A \cup B$ is shaded

Theorem

Let A and B be sets. Then

1. $A \subseteq A \cup B$ *and* $B \subseteq A \cup B$,
2. $A \subseteq B \Leftrightarrow A \cup B = B$.

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Proof.

1. Let $x \in A$, then

$$\begin{aligned} x \in A &\Rightarrow x \in A \vee x \in B \\ &\Rightarrow x \in A \cup B. \end{aligned}$$

Thus $A \subseteq A \cup B$. By the same manner, we prove that $B \subseteq A \cup B$

2. Suppose $A \subseteq B$ and $x \in A \cup B$. Then

$$\begin{aligned} x \in A \cup B &\Rightarrow x \in A \vee x \in B \\ &\Rightarrow x \in B \vee x \in B \ \ (\textit{since} \ \ A \subseteq B) \\ &\Rightarrow x \in B. \end{aligned}$$

Thus $A \cup B \subseteq B$. Since $B \subseteq A \cup B$ (by part (1)), then $A \cup B = B$. Conversely, suppose that $A \cup B = B$. Since $A \subseteq A \cup B$ (by part (1)), then $A \subseteq B$.



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Proposition

Let A, B, and C be sets. Then

1. $A \cup A = A$ *(Idempotent law)*,
2. $A \cup B = B \cup A$ *(Commutative law)*,
3. $A \cup (B \cup C) = (A \cup B) \cup C$ *(Associative law)*.
4. $A \cup \emptyset = A$,
5. $A \cup E = E$, *where E is the universal set.*

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Intersection

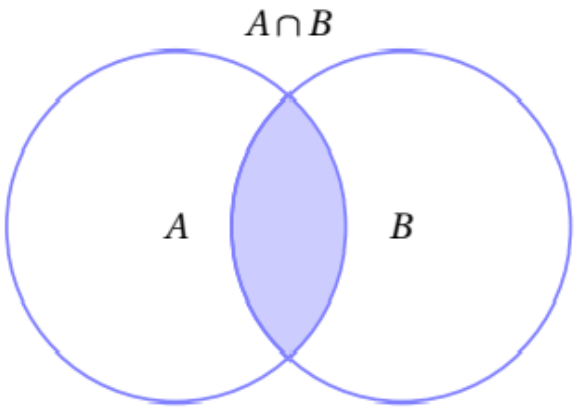
Definition

If A and B are sets. **The intersection** of A and B, denoted by $A \cap B$, is defined by

$$A \cap B = \{x \in E \mid (x \in A) \wedge (x \in B)\}$$

That is,

$$x \in A \cap B \Leftrightarrow x \in A \wedge x \in B.$$

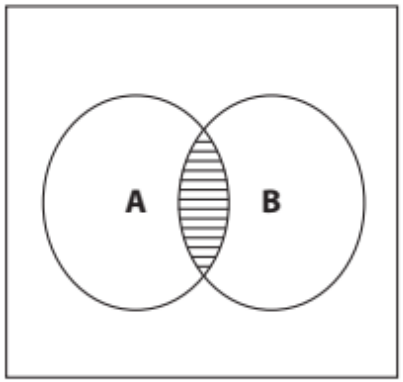


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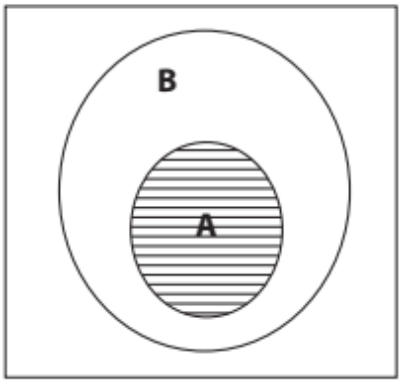
Example

Let $A = \{x \in \mathbb{N} : x \leq 6\}$,
$B = \{x : x \ \ is \ \ a \ \ prime \ \ number \ \ and \ \ \leq 6\}$. Then

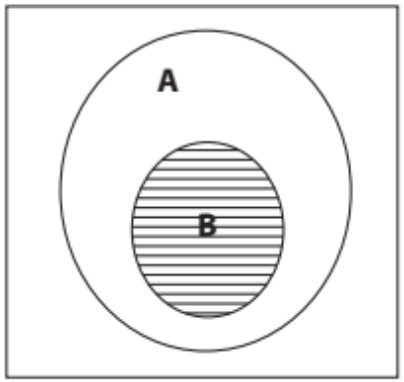
$$A \cap B = \{1, 2, 3, 4, 5, 6\} \cap \{2, 3, 5\} = \{2, 3, 5\}$$



$A \cap B$ is shaded



$A \cap B$ is shaded



$A \cap B$ is shaded

Definition

Let A, B be sets. We say that A and B are **disjoint** iff

$$A \cap B = \emptyset$$

Example

Let $A = \{x \in \mathbb{N} : x \ \ is \ \ an \ \ odd \ \ number\}$ and $B = \{x \in \mathbb{N} : x \ \ is \ \ an \ \ even \ \ number\}$.
Since $A \cap B = \emptyset$, then A and B are disjoint. But $A \cap \mathbb{N} \neq \emptyset$, thus A and $\mathbb{N}$ are not disjoint.

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Proposition

Let A and B be sets. Then

1. $A \cap B \subseteq A$ *and* $A \cap B \subseteq B$,
2. $A \subseteq B \Leftrightarrow A \cap B = A$.

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Theorem

Let A, B, and C be sets. Then

1. $A \cap A = A$ *(Idempotent law)*,
2. $A \cap B = B \cap A$ *(Commutative law)*,
3. $A \cap (B \cap C) = (A \cap B) \cap C$ *(Associative law)*.
4. $A \cap \emptyset = \emptyset$,
5. $A \cap E = A$, *where E is the universal set.*

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Proof.

We prove (c) only. Let $x \in A \cap (B \cap C)$. Then

$$\begin{aligned} x \in A \cap (B \cap C) &\Leftrightarrow x \in A \wedge x \in (B \cap C) \\ &\Leftrightarrow x \in A \wedge (x \in B \wedge x \in C) \\ &\Leftrightarrow (x \in A \wedge x \in B) \wedge x \in C \\ &\Leftrightarrow x \in (A \cap B) \wedge x \in C \\ &\Leftrightarrow x \in (A \cap B) \cap C. \end{aligned}$$

Thus $A \cap (B \cap C) = (A \cap B) \cap C$. □

Theorem (Distributive law)

Let A,B, and C be sets. Then

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Theorem

Let A and B be sets. Then

1. $A \subseteq B \Leftrightarrow P(A) \subseteq P(B)$,
2. $P(A) \cap P(B) = P(A \cap B)$,
3. $P(A) \cup P(B) \supseteq P(A \cup B)$.

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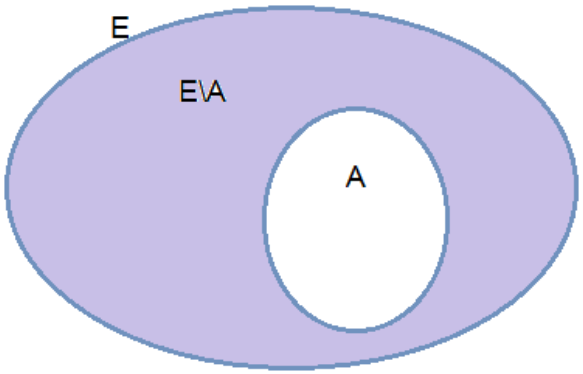
COMPLEMENT OF SET

Definition

Let E be the universal set and $A \subseteq E$.
The set

$$E \setminus A = \{x \in E \,|\, x \notin A\} = A^C = \overline{A}.$$

is said to be ***the complement of*** A.



Example

Let the universal set E be $\mathbb{N}$ and let

$$A = \{2, 4, 6, 8, ...\}.$$

Then

$$A^C = \{1, 3, 5, 7, ...\}.$$

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Theorem

Let A and B be sets. Then

$$A \subseteq B \Rightarrow B^C \subseteq A^C.$$

Proof.

Since $A \subseteq B$, then $x \in A \Rightarrow x \in B$.
Thus $x \notin B \Rightarrow x \notin A$.
That is, $x \in B^C \Rightarrow x \in A^C$.
Hence $B^C \subseteq A^C$. □

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Proposition

Let A be a set. Then

$$(A^C)^C = A.$$

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Proposition

Let A be a set. Then

$$(A^C)^C = A.$$

Proof.

It is clear that

$$\begin{aligned} x \in (A^C)^C &\Leftrightarrow x \notin A^C \\ &\Rightarrow x \in A. \end{aligned}$$

Thus $(A^C)^C = A$.

□

Theorem

Let A be a set. Then

1. $E^C = \emptyset$,
2. $\emptyset^C = E$,
3. $A \cap A^C = \emptyset$,
4. $A \cup A^C = E$.

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Theorem

Let A and B be sets. Then

1. $(A \cup B)^C = A^C \cap B^C$
2. $(A \cap B)^C = A^C \cup B^C$,

Proof [1: Worksheet N°:10] Let A and B be two subsets of E. Show that:

2. $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

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Let A and B be two subsets of a set E. We want to show that:

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Proof that $(A \cup B)^c = A^c \cap B^c$
Let $x \in E$.

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff \text{not}(x \in A \text{ or } x \in B) \\ &\iff (x \notin A) \text{ and } (x \notin B) \\ &\iff x \in A^c \text{ and } x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

Hence, $(A \cup B)^c = A^c \cap B^c$.

Proof that $(A \cap B)^c = A^c \cup B^c$

Let $x \in E$.

$$\begin{aligned} x \in (A \cap B)^c &\iff x \notin A \cap B \\ &\iff \text{not}(x \in A \text{ and } x \in B) \\ &\iff (x \notin A) \text{ or } (x \notin B) \\ &\iff x \in A^c \text{ or } x \in B^c \\ &\iff x \in A^c \cup B^c. \end{aligned}$$

Thus, $(A \cap B)^c = A^c \cup B^c$.

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The difference of sets

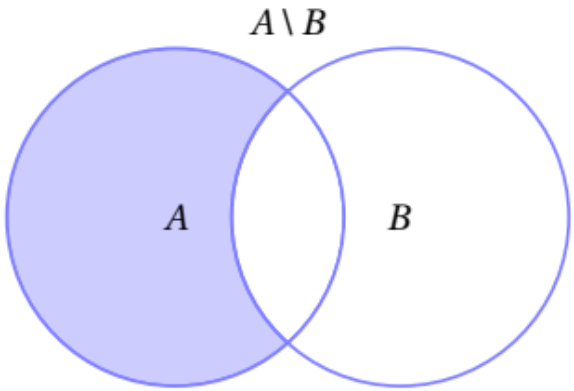
Definition

Let A and B be sets. The difference of A and B is defined by

$$A\backslash B = \{x \in E \,|\, ((x \in A) \ \textit{and} \ (x \notin B))\} = A \cap (E \backslash B).$$

Or

$$A - B = \{x, x \in A \ \ \textit{and} \ \ x \notin B\}.$$



Remark

$$A \setminus B = A \cap B^C.$$

Let A and B be sets. Then

$$A - B = B^C - A^C.$$

Proof.

We have

$$\begin{aligned} x \in A - B &\Leftrightarrow x \in A \wedge x \notin B \\ &\Leftrightarrow x \notin A^C \wedge x \in B^C \\ &\Leftrightarrow x \in B^C - A^C. \end{aligned}$$

Thus $A - B = B^C - A^C$. □

Generalized union and intersection

Definition

Let $F_1, \ldots F_i, \ldots$ be an indexed family of sets. Then

$$\bigcap_{i \in \mathbb{N}} F_i = \{x \in E \mid (\forall i \in \mathbb{N})(x \in F_i)\},$$

and

$$\bigcup_{i \in \mathbb{N}} F_i = \{x \in E \mid (\exists i \in \mathbb{N})(x \in F_i)\}.$$

Example

Let $B_n = [0, 1/n]$, where $n \in \mathbb{N}^*$. Then

$$\bigcup_{i \in \mathbb{N}} B_i = [0, 1],$$

and

$$\bigcap_{i \in \mathbb{N}} B_i = \{0\}.$$

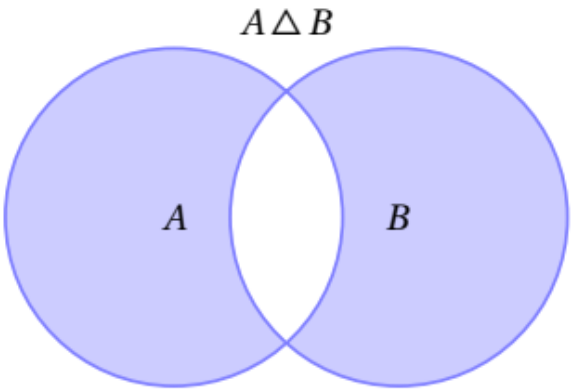
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Symmetric Difference of Sets

Definition

Let A and B be sets. **The symmetric difference** of A and B is defined by

$$A\Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$



Theorem

Let A and B be sets. Then

1. $A\Delta\emptyset = A$,
2. $A\Delta B = \emptyset \Leftrightarrow A = B$.

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Cartesian Product

Definition

Let A and B be sets. The Cartesian product of A and B consists of all ordered pairs (a, b), where $a \in A$ and $b \in B$. It is denoted by

$$A \times B,$$

which reads "A **cross** B". More concisely,

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Remark

In general we have

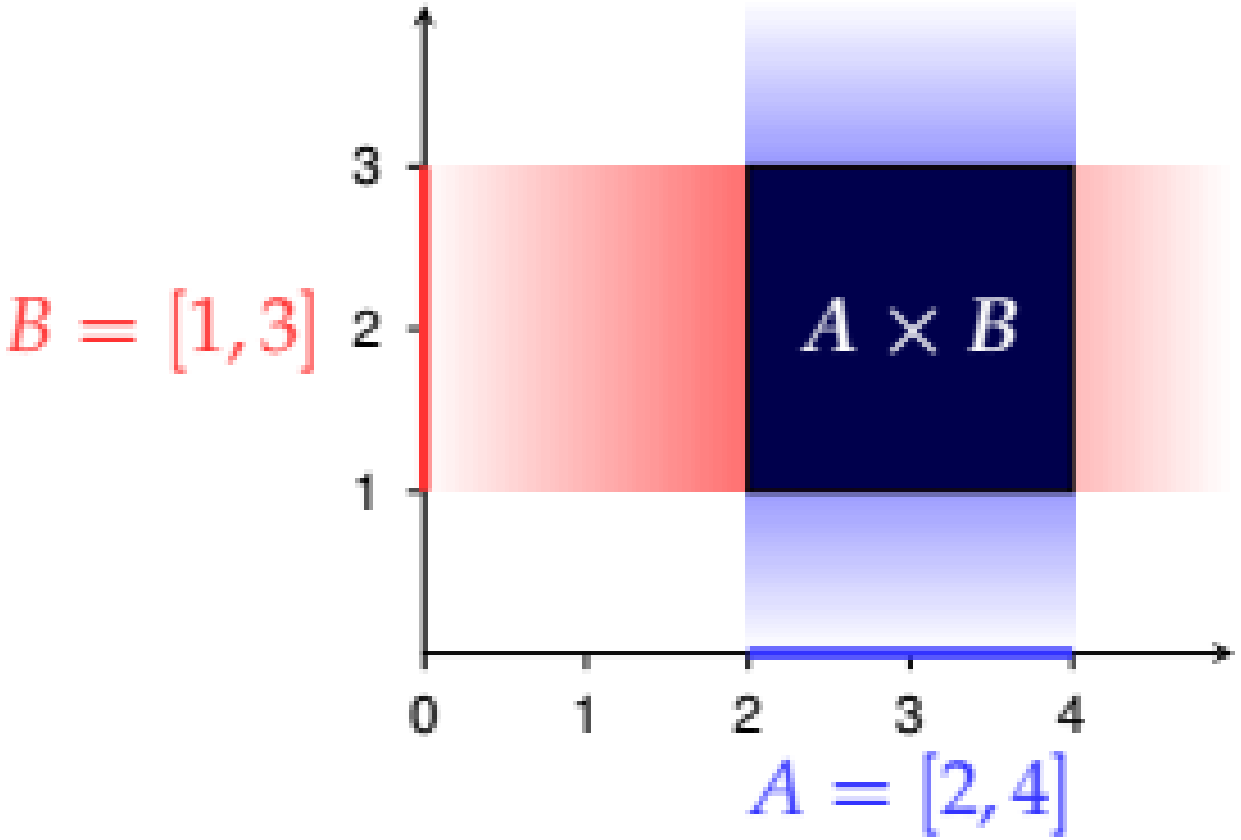
$$A \times B \neq B \times A.$$

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The difference of sets
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Cartesian Product

Example

The Cartesian product of $A = [2, 4]$ and $B = [1, 3]$ is

$$A \times B = \{(x, y) | x \in [2, 4] \quad \textit{and} \quad y \in [1, 3]\}$$



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Example

1. Let $C = \{x, y\}$. Then

$$C \times C = \{(x, x), (x, y), (y, x), (y, y)\}.$$

2. Let $A = \mathbb{R}$ the set of real numbers. Then

$$\mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\} = \mathbb{R}^2.$$

The elements of $\mathbb{R} \times \mathbb{R}$ are the points of the $xy - plane$ (Cartesian plane).

Example

The Cartesian product of $A = \{0, 1\}$ and $B = \{2, 3, 4\}$ is

$$A \times B = \{(0, 2), (0, 3), (0, 4), (1, 2), (1, 3), (1, 4)\}.$$

We have

$A \times B$	2	3	4
0	(0,2)	(0,3)	(0,4)
1	(1,2)	(1,3)	(1,4)